



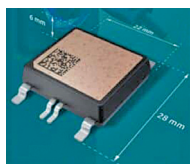
NEW PRODUCTS

MAKE IN INDIA

COMPONENTS & CHIPS

A SiC rectifier module

Nexperia, in collaboration with KYOCERA AVX, has launched a 650V, 20A silicon carbide (SiC) rectifier module. The module is optimised for high-frequency power operations ranging between 3kW to 11kW power



stack designs. Target applications encompass industrial power supplies, EV charging stations, and

on-board chargers. The module also optimises thermal performance through top-side cooling and incorporates an integrated negative temperature coefficient (NTC) sensor, providing continuous temperature monitoring for real-time feedback crucial for accurate prognosis and diagnosis at both the device and system levels. Additionally, its low inductance package enables high-frequency operations, and it holds certification for operation with a junction temperature as high as 175°C. The combination of silicon carbide semiconductors with advanced module packaging enhances the company's ability to meet the growing demand for power electronic products with superior power density.

Nexperia

<https://www.nexperia.com>

8-megapixel CMOS image sensor

OMNIVISION has launched its OX08D10 8-megapixel (8MP) CMOS image sensor featuring TheiaCel technology. The company introduces an innovative solution aimed at enhanc-

ing automotive safety by delivering superior resolution and image quality for exterior cameras used in advanced driver assistance systems (ADAS) and autonomous driving. The TheiaCel technology empowers the sensor to excel in high dynamic range (HDR)



image capture, extending its effectiveness up to 200 metres, a range finely balancing signal-to-noise ratio

(SNR) and dynamic range, making it a top choice for automotive exterior camera applications. Noteworthy features include exceptional low-light performance, low power consumption, and a compact form factor that's 50% smaller than similar exterior cabin sensors. It's the first image sensor to utilise next-generation lateral overflow integration capacitors alongside HDR technology, eliminating LED flicker across varying lighting conditions while maintaining robust light flicker mitigation capabilities without compromising image quality.

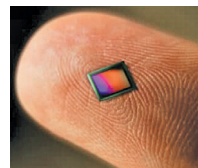
OMNIVISION

<https://www.ovt.com>

Lowest power image sensors

Onsemi has introduced a 1.4µm pixel image sensor family, Hyperlux LP. This product family adopts a stacked architecture design that minimises its footprint, reaching a size similar to a grain of rice in its smallest form. The image sensors enhance camera performance, providing clearer images and more accurate object detection even in challenging weather and lighting

conditions. The sensors incorporate a wake-on-motion feature, allowing them to operate in a low-power mode until movement is detected. At this point, they transition to a higher-



performance state. Smart region of interest (RoI) management also enables multiple RoIs to offer con-

text and detailed views while conserving bandwidth. The versatile image-sensing family finds applications in industrial and commercial cameras, including smart doorbells, security cameras, augmented reality/virtual reality/extended reality (AR/VR/XR) headsets, machine vision, and video conferencing.

Onsemi

<https://www.onsemi.com>

Compact open-frame AC/DC converters

The RECOM RACM30-K/277 series of budget-friendly board-mount AC/DC converters has been expanded to offer an open-frame alternative, reducing the weight by half compared to the existing encapsulated variants.



Safety certifications cover a wide range of applications, including audio/video, informa-

tion and communication technology (ICT), test equipment, household appliances, industrial devices, and medical equipment, with robust 4kV AC/1-minute-rated isolation. These converters also comply with eco-design requirements, exhibiting minimal no-load/standby power dissipation and maintaining nearly flat efficiency